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Electronics Basic

6-1

Bipolar Junction Transistors



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Outcome

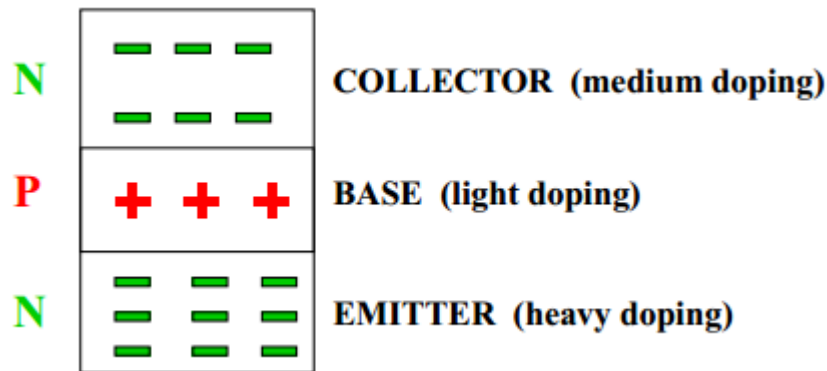
- **Unbiased transistor**
- **Biased transistor**
- **Transistor currents**
- **The CE connection**
- **The base curve**
- **Collector curves**
- **Transistor approximations**
- **Reading data sheets**
- **Surface mount transistors**
- **Troubleshooting**



Unbiased transistor

- Three doped regions: emitter, base, and collector
- Two pn junctions: emitter-base and base-collector
- NPN or PNP
- Silicon or germanium

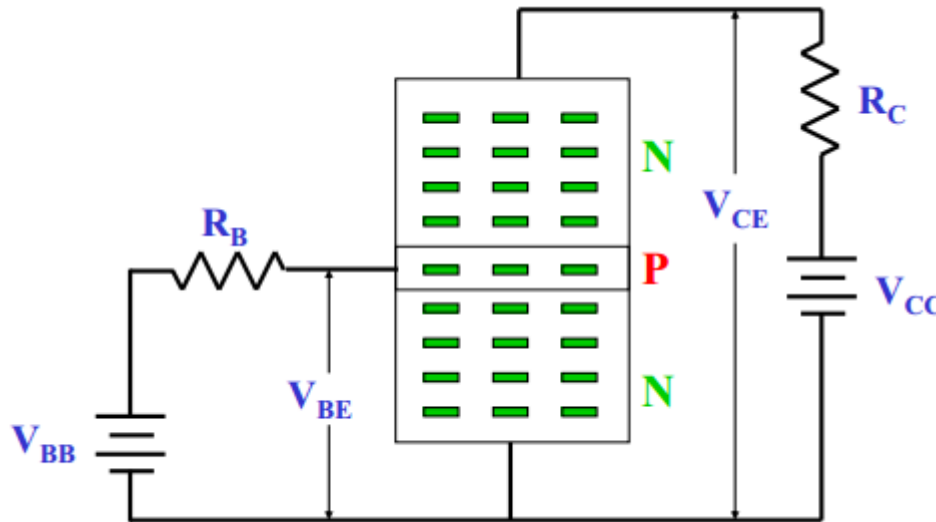
The bipolar junction transistor has 3 doped regions.



Biased transistor

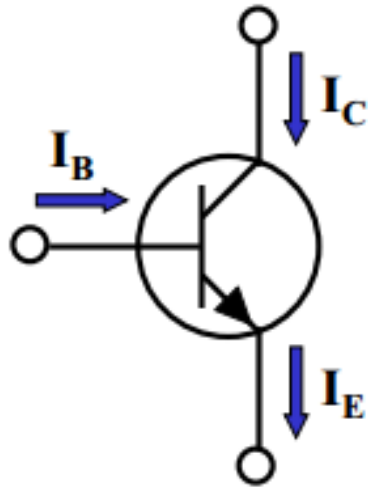
- Forward bias the **emitter** diode
- Reverse bias the **collector** diode

In a properly biased NPN transistor, the **emitter** electrons diffuse into the base and then go on to the collector.



Transistor currents

- The **ratio** of collector current to base current is **current gain** (β_{dc} or h_{FE})
- Current gain is typically **100 to 300**



Conventional flow

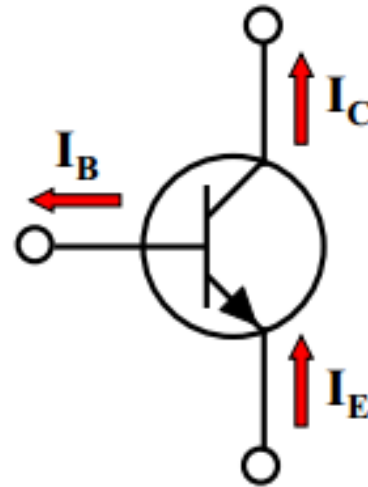
$$I_E = I_C + I_B$$

$$I_C \cong I_E$$

$$I_B \ll I_C$$

$$\alpha_{dc} = \frac{I_C}{I_E}$$

$$\beta_{dc} = \frac{I_C}{I_B}$$



Electron flow

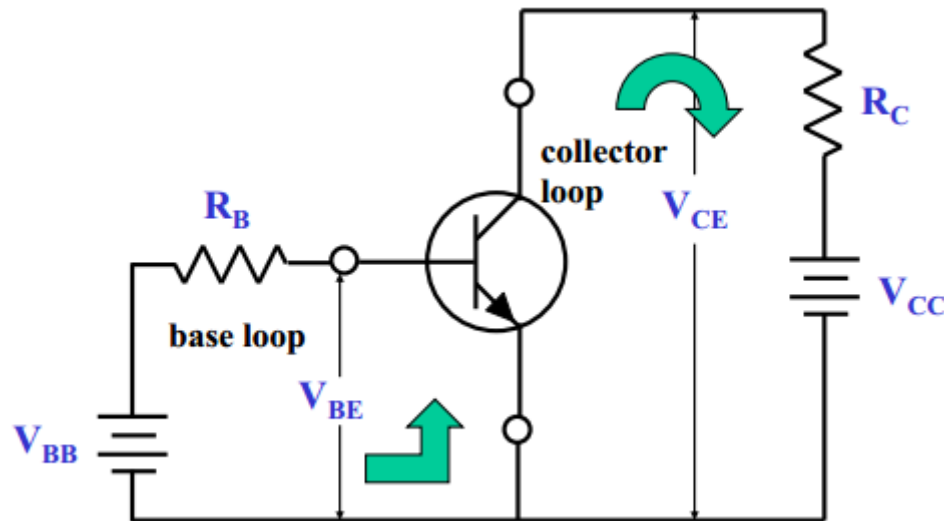


The CE connection

- The emitter is grounded or common
- The base-emitter acts like a diode
- The base-collector acts like a current

source that is equal to β_{dc} times the base current

The common emitter connection has two loops: the base loop and the collector loop.



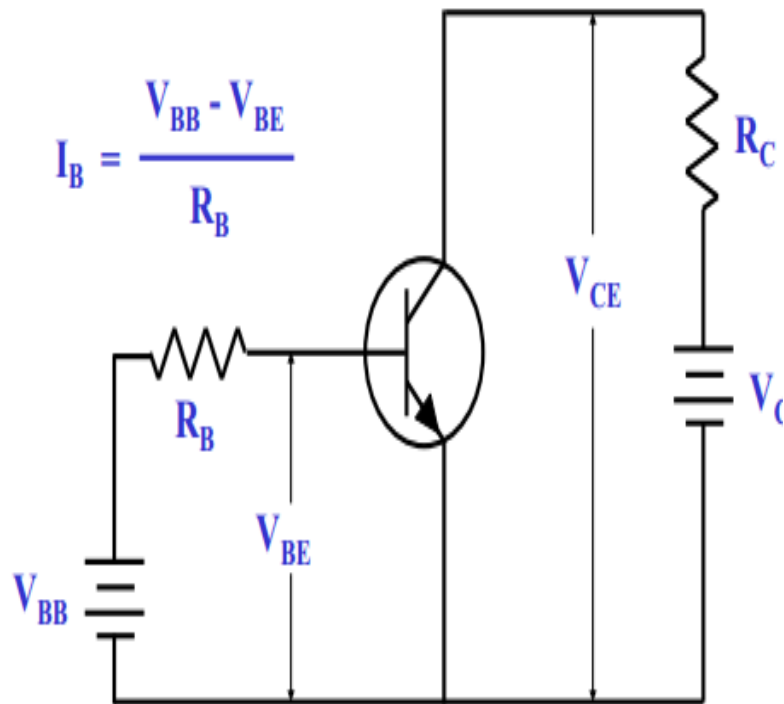
Subscript notation

- When the subscripts are the same, the voltage represents a source (VCC).
- When the subscripts are different, the voltage is between two points (VCE).
- Single subscripts are used for node voltages with ground serving as the reference (VC).

Base curve

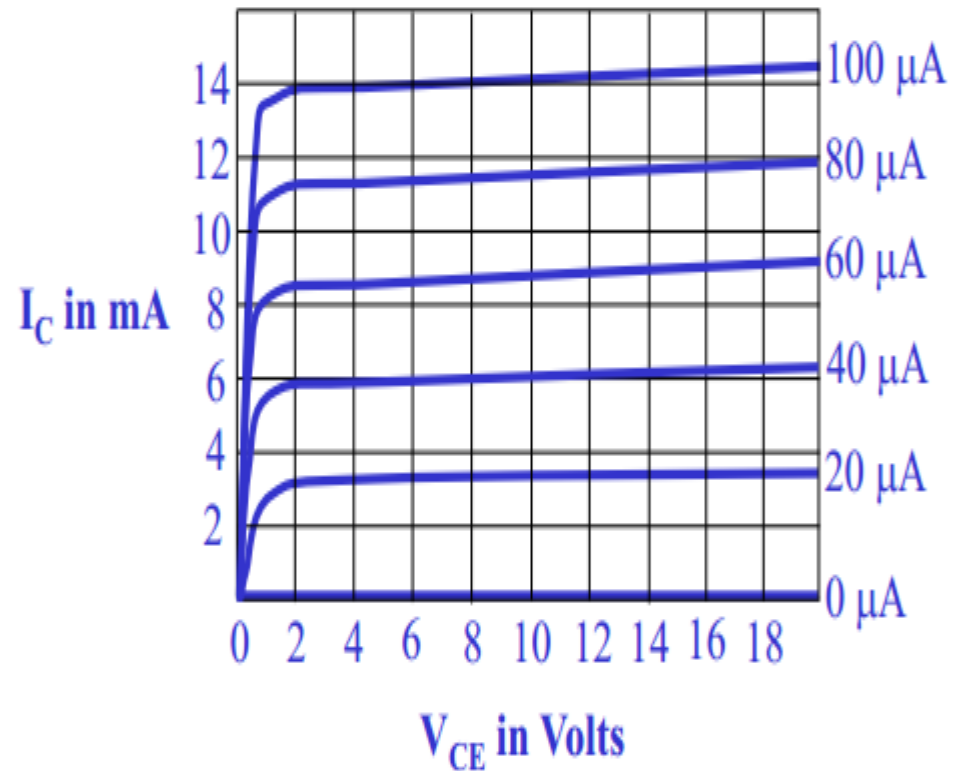
- Graph similar to that of a diode
- Diode approximations are used for analysis (typically - ideal or second)

The base circuit is usually analyzed with the same approximation used for diodes.



A graph of I_C versus V_{CE}

(Note that each new value of I_B presents a new curve.)



This set of curves is also called a family of curves



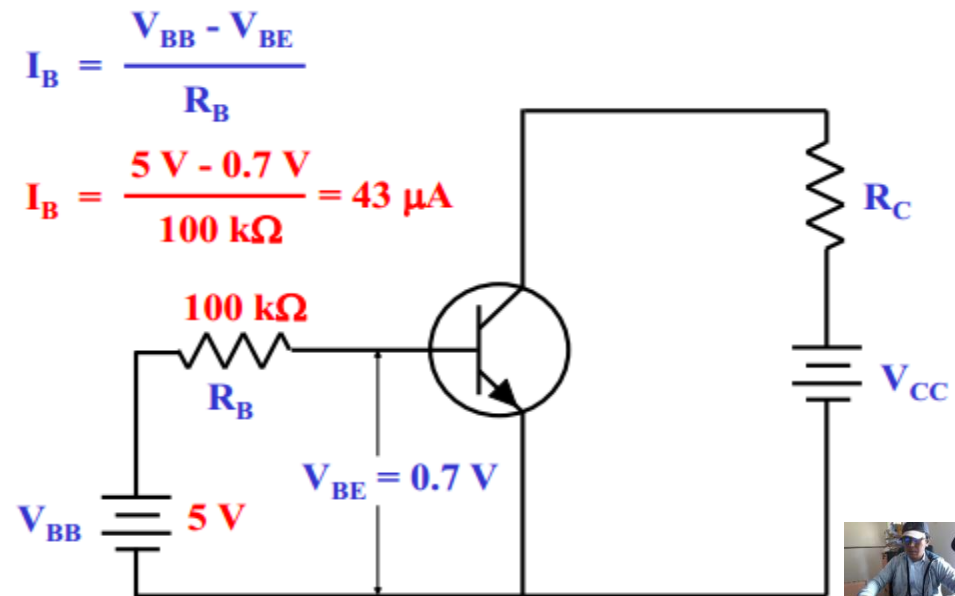
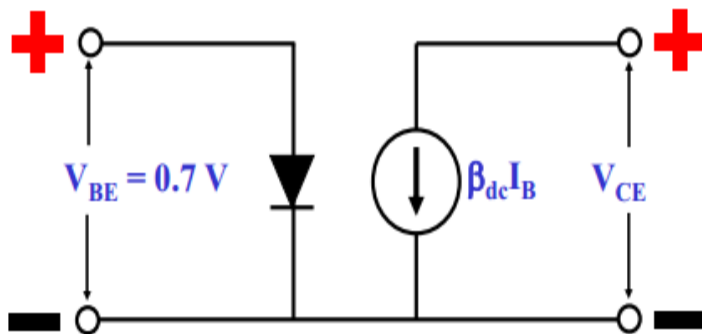
Regions of operation

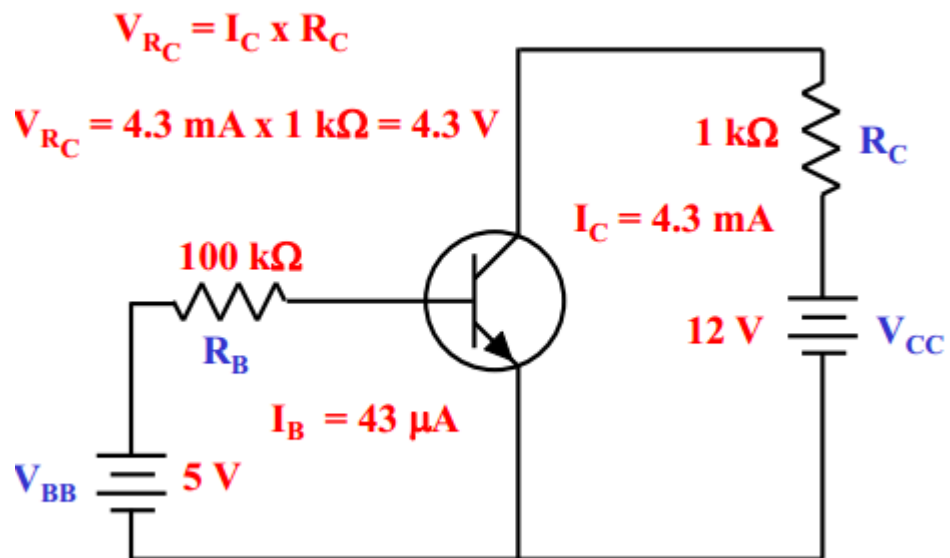
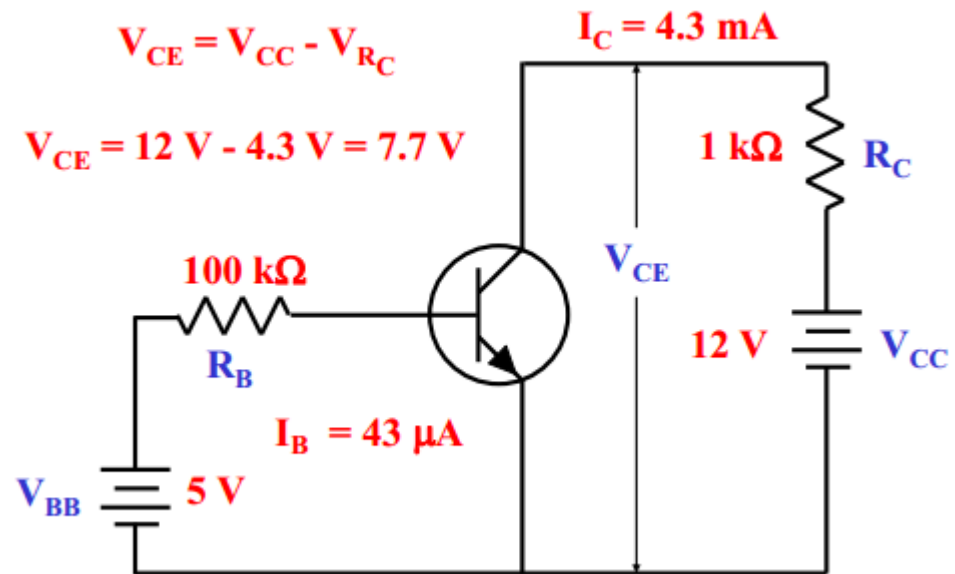
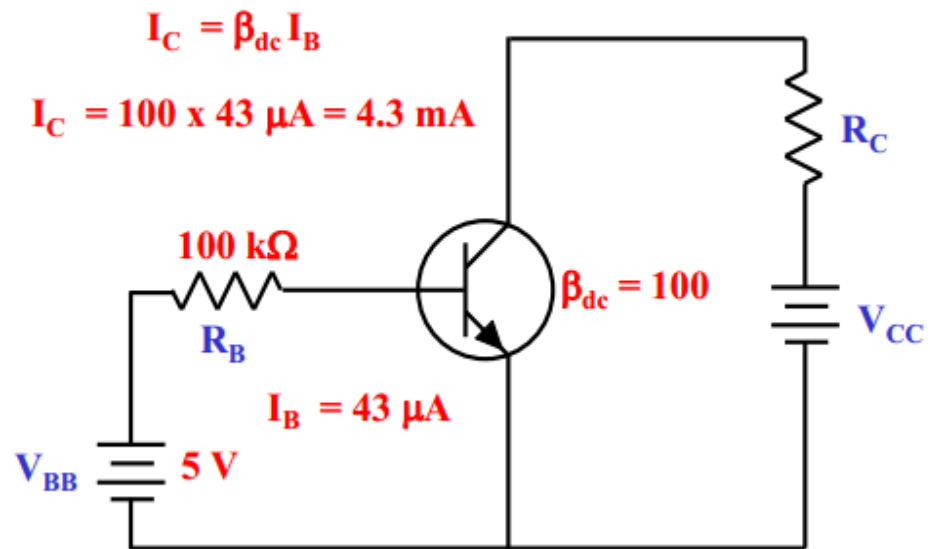
- **Active** - - - used for linear amplification
- **Cutoff** - - - used in switching applications
- **Saturation** - - - used in switching applications
- **Breakdown** - - - can destroy the transistor and should be avoided

Transistor circuit approximations

- **First:** treat the base-emitter diode as ideal and use βI_B to determine I_C . Use for troubleshooting.
- **Second:** correct for V_{BE} and use βI_B to determine I_C .
- **Third (and higher):** correct for bulk resistance and other effects. Usually accomplished by computer simulation. Use for design work.

The second approximation:





Reading transistor data sheets

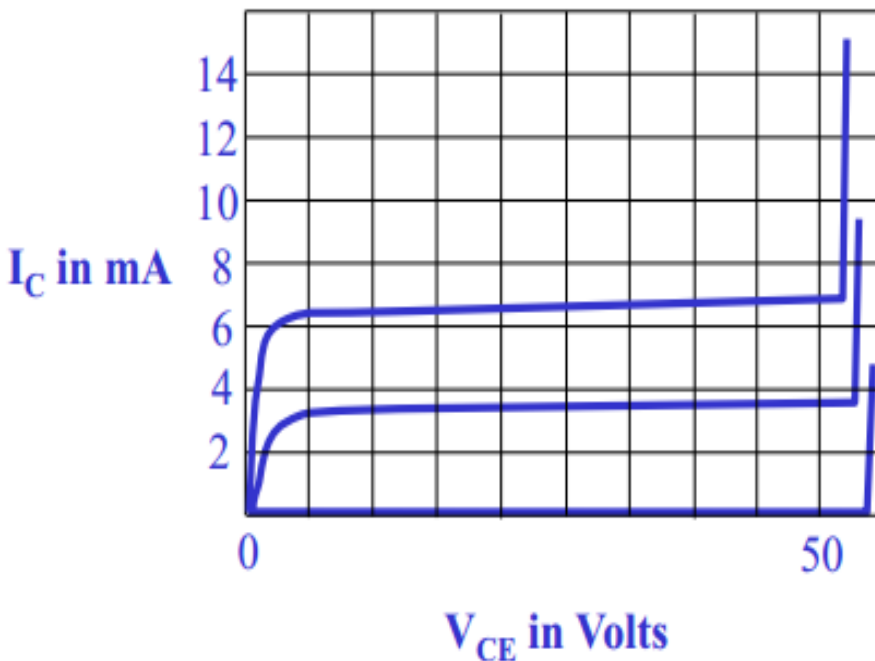
- Maximum ratings on voltage, current, and power
- Power transistors dissipate more than 1 watt
- Temperature can change the value of a transistor's characteristics

Typical Breakdown Ratings

- $V_{CBO} = 60 \text{ V}$
- $V_{CEO} = 40 \text{ V}$
- $V_{EBO} = 6 \text{ V}$
- Note: these are reverse breakdown ratings with one transistor leg open (e.g. V_{CBO} is voltage collector to base with emitter open)



A graphic view of collector breakdown



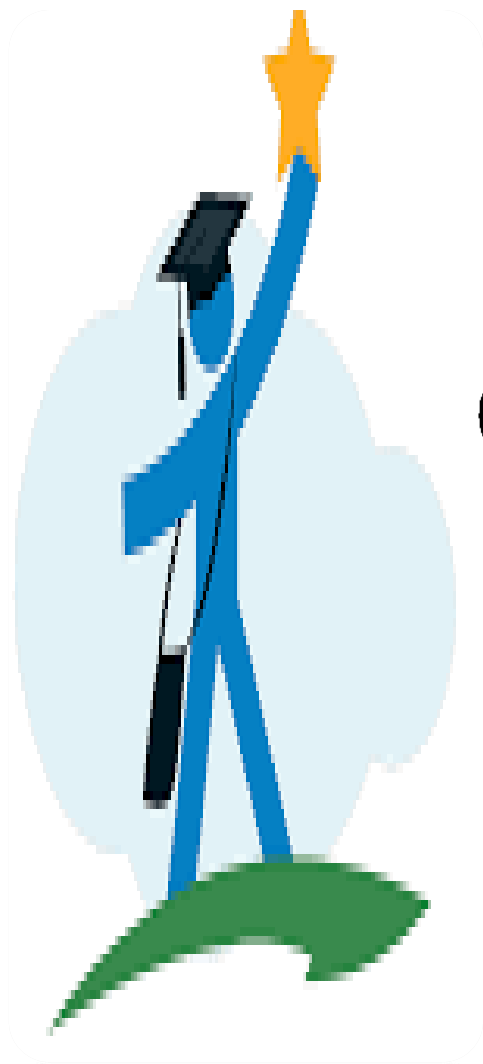
Typical Maximum Ratings

- $I_C = 200$ mA dc
- $P_D = 250$ mW (for $T_A = 60$ °C)
- $P_D = 350$ mW (for $T_A = 25$ °C)
- $P_D = 1$ W (for $T_C = 60$ °C)

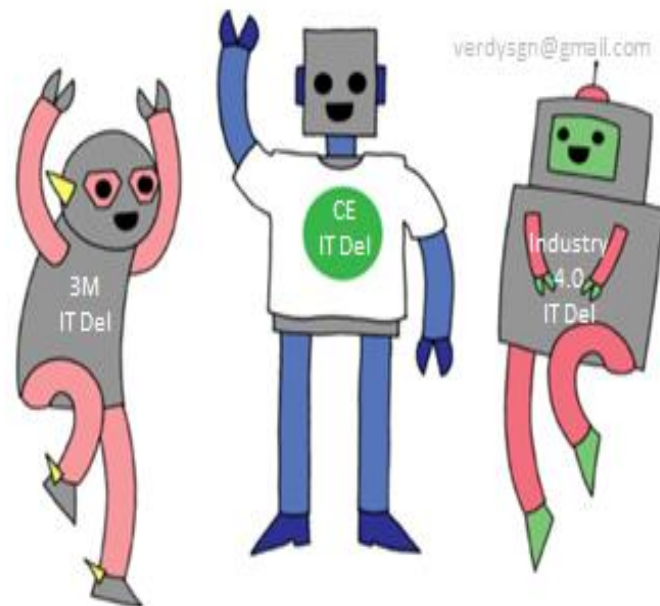
Data sheet h_{FE} “On Characteristics”

I_C in mA	$h_{FE(min)}$	$h_{FE(max)}$
0.1	40	—
1	70	—
10	100	300
50	60	—
100	30	—





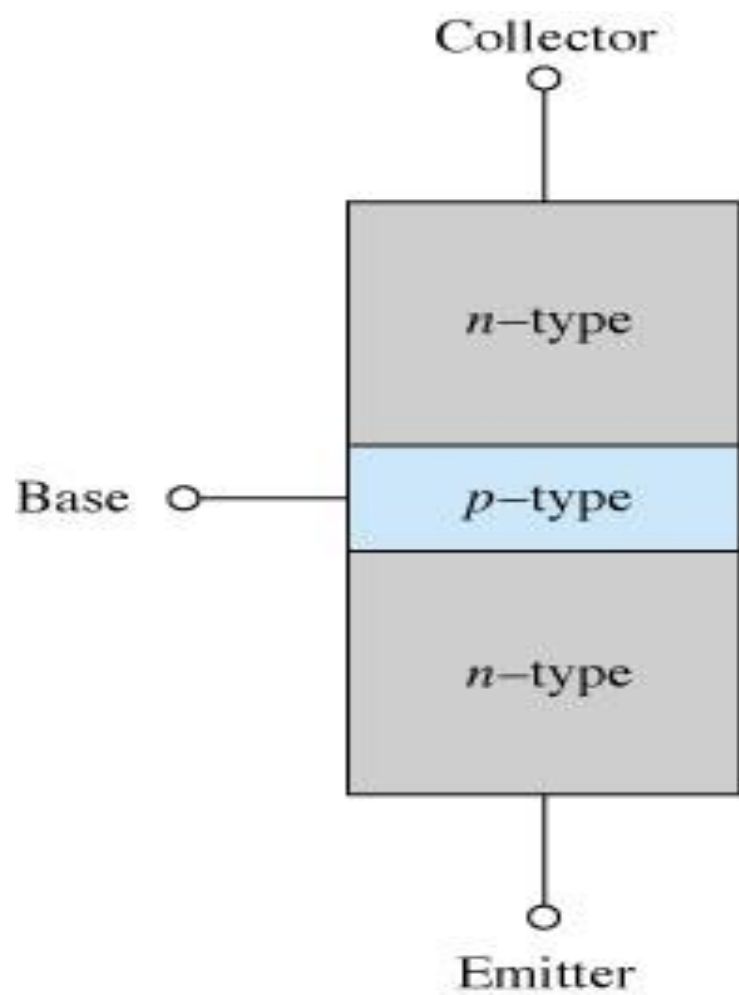
Thanks!



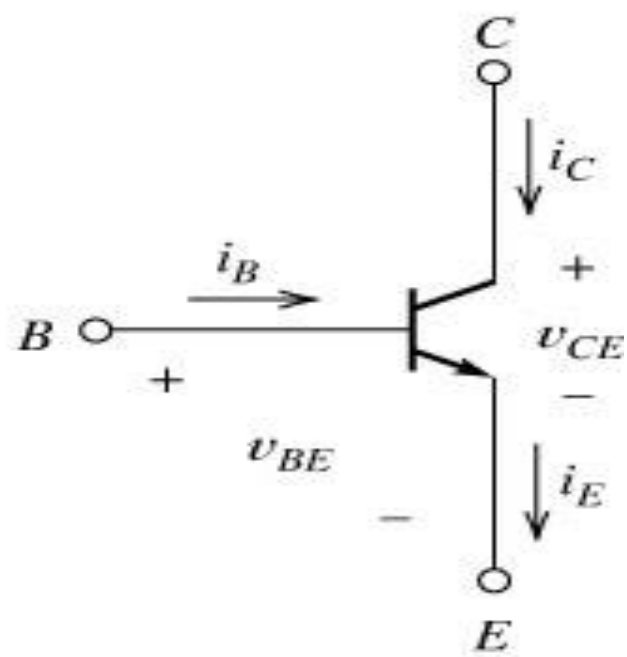
Transistor

- Three doped regions
- Emitter, Base and Collector
- Base region is much thinner as compared to the collector and emitter
- ***npn*** and ***pnp***
- Emitter is heavily doped, Base is lightly and collector is intermediate
- Collector regions is physically largest

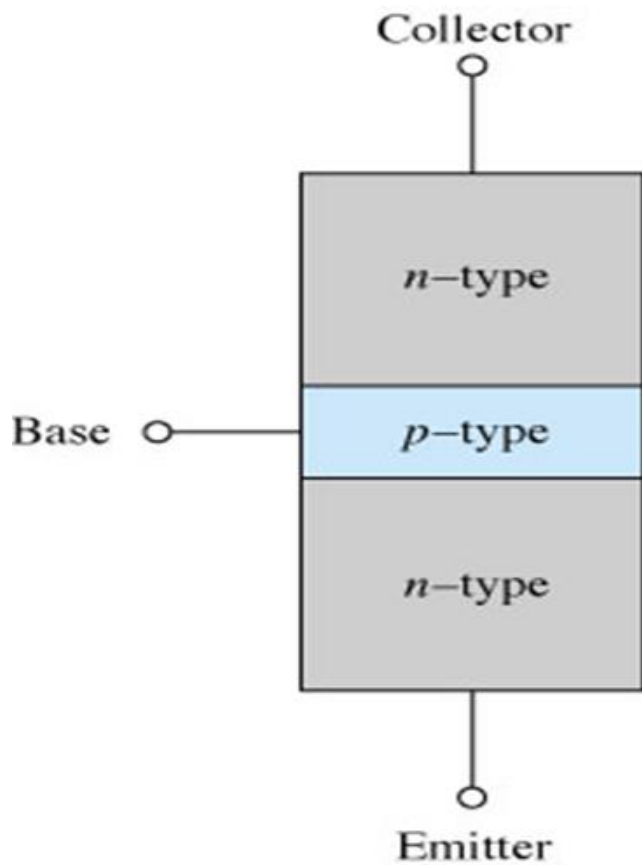




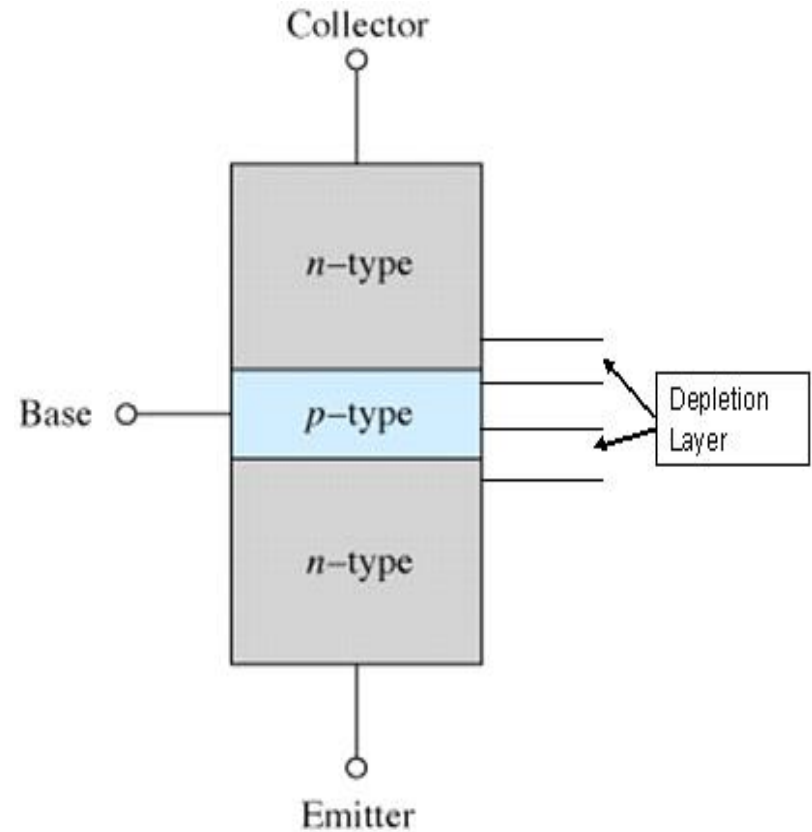
(a) Simplified physical structure



(b) Circuit symbol



Before Diffusion



After Diffusion

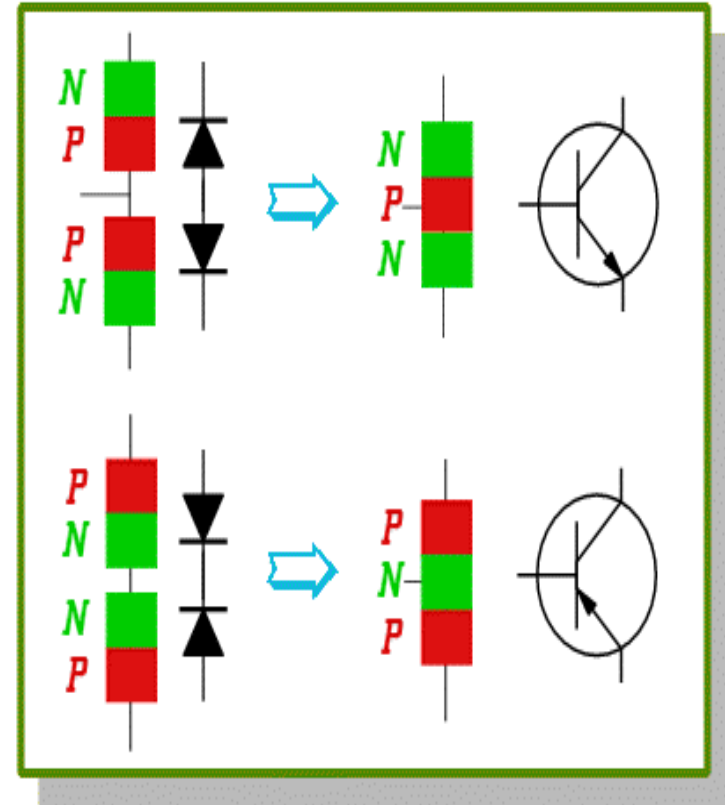
Each of Dep. Layer barrier potential app. 0.7 V at 25° C

Unbiased transistor is like two back-to-back diodes

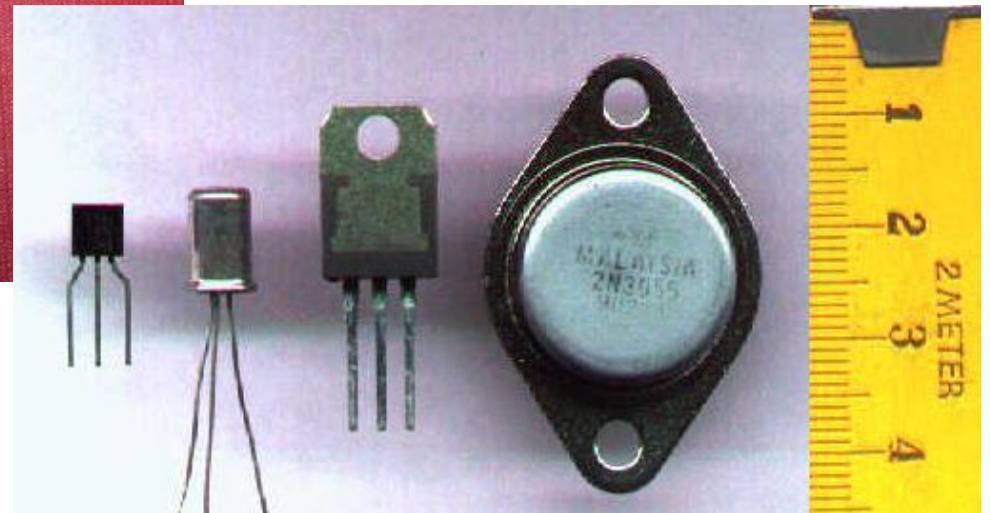


Bipolar Junction Transistors

- A bipolar transistor essentially consists of a pair of **PN Junction diodes** that are joined back-to-back.
- There are therefore two kinds of BJT, the **NPN** and **PNP** varieties.
- The three layers of the sandwich are conventionally called the **Collector**, **Base**, and **Emitter**.

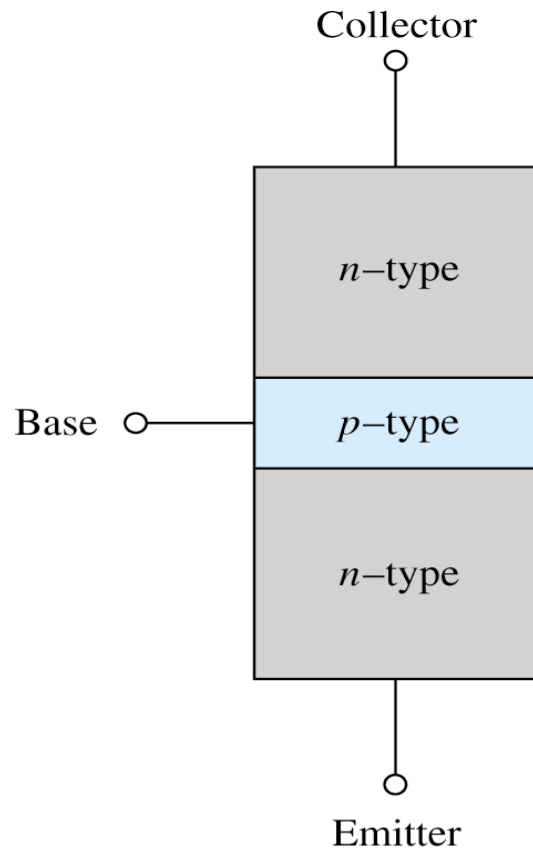


Modern Transistors

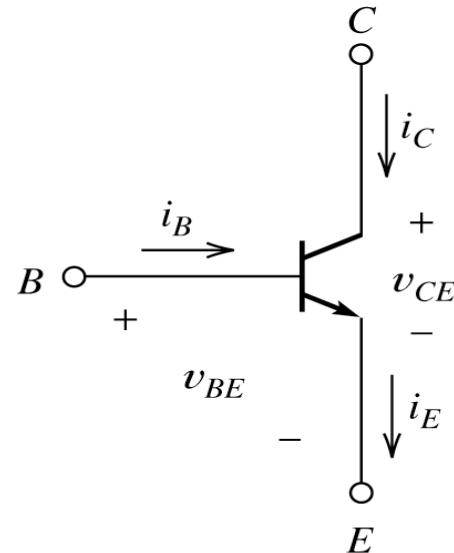


NPN Bipolar Junction Transistor

- One N-P (Base Collector) diode one P-N (Base Emitter) diode



(a) Physical structure



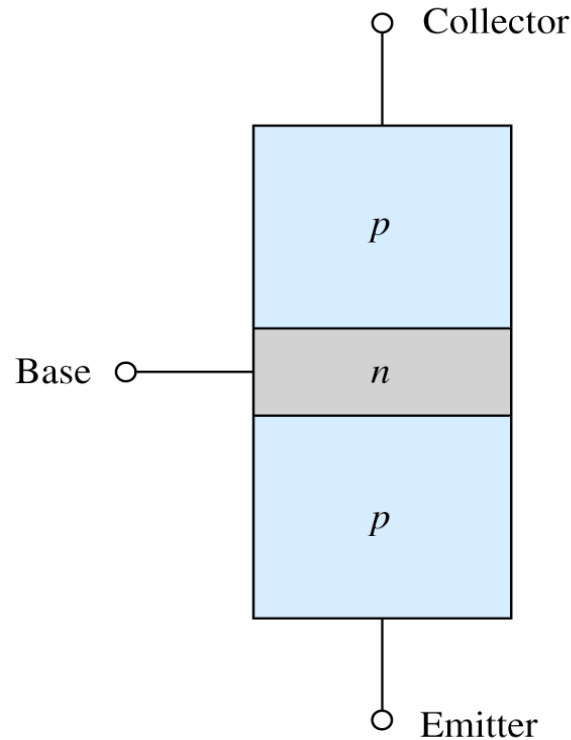
(b) Circuit symbol

Figure 13.1 The *npn* BJT.

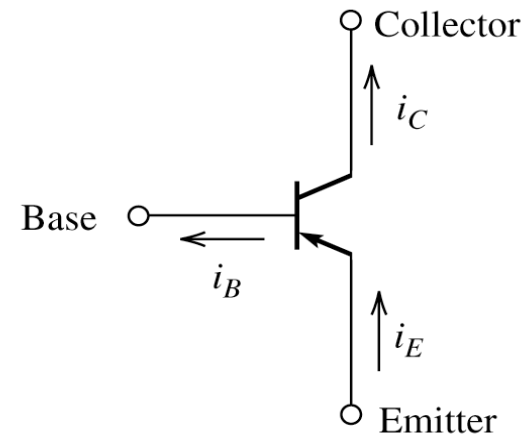


PNP Bipolar Junction Transistor

- One P-N (Base Collector) diode one N-P (Base Emitter) diode



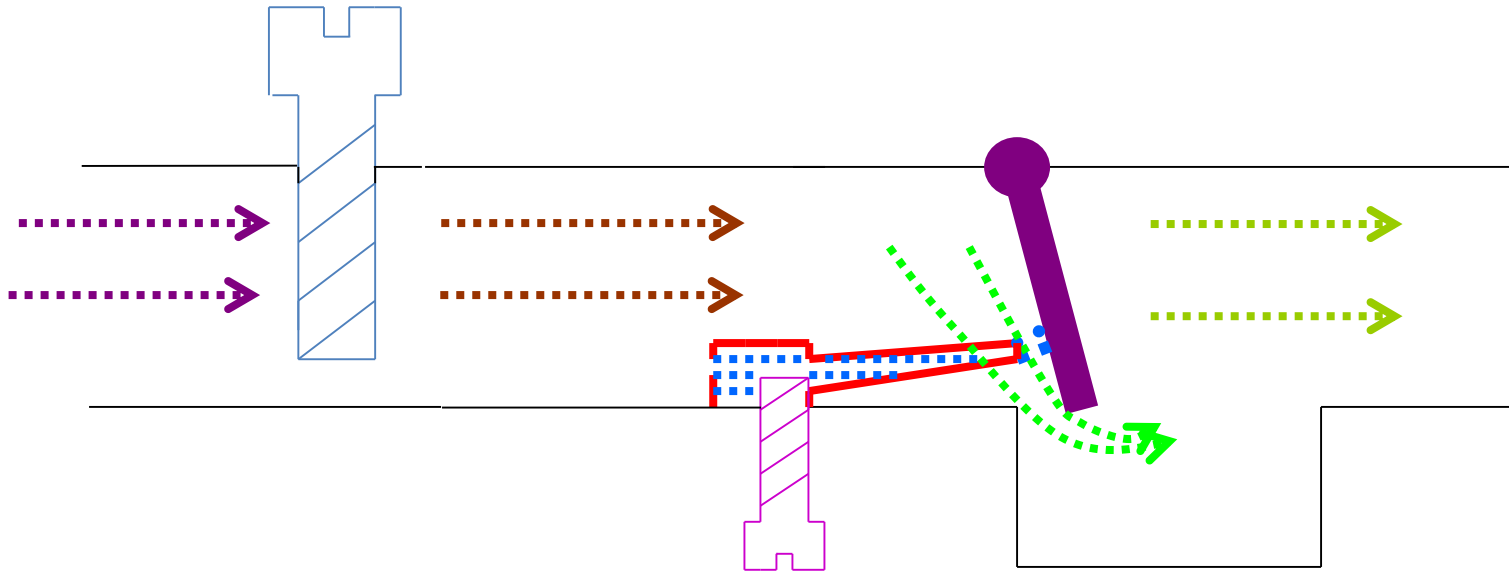
(a) Physical structure



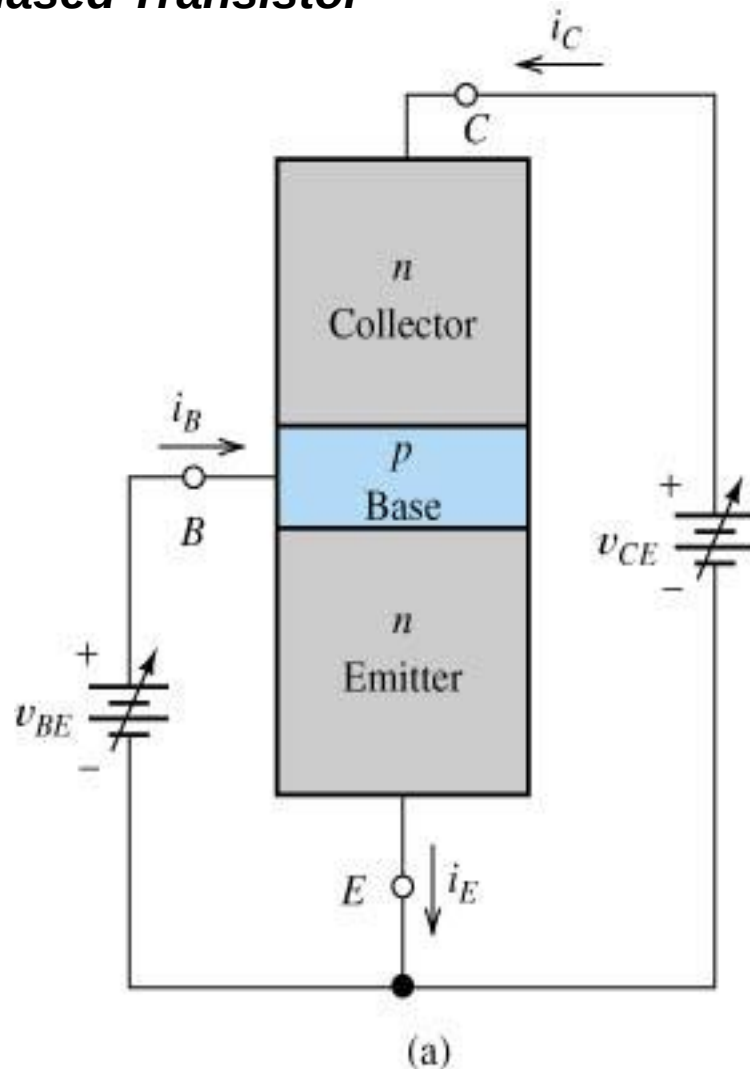
(b) Circuit symbol with reference directions for currents

Figure 13.13 The *pn*p BJT.

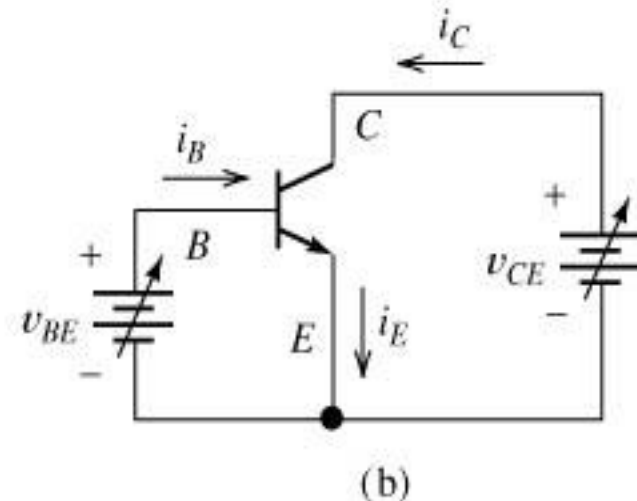
Analogy with Transistor :Fluid-jet operated Valve



****The Biased Transistor***



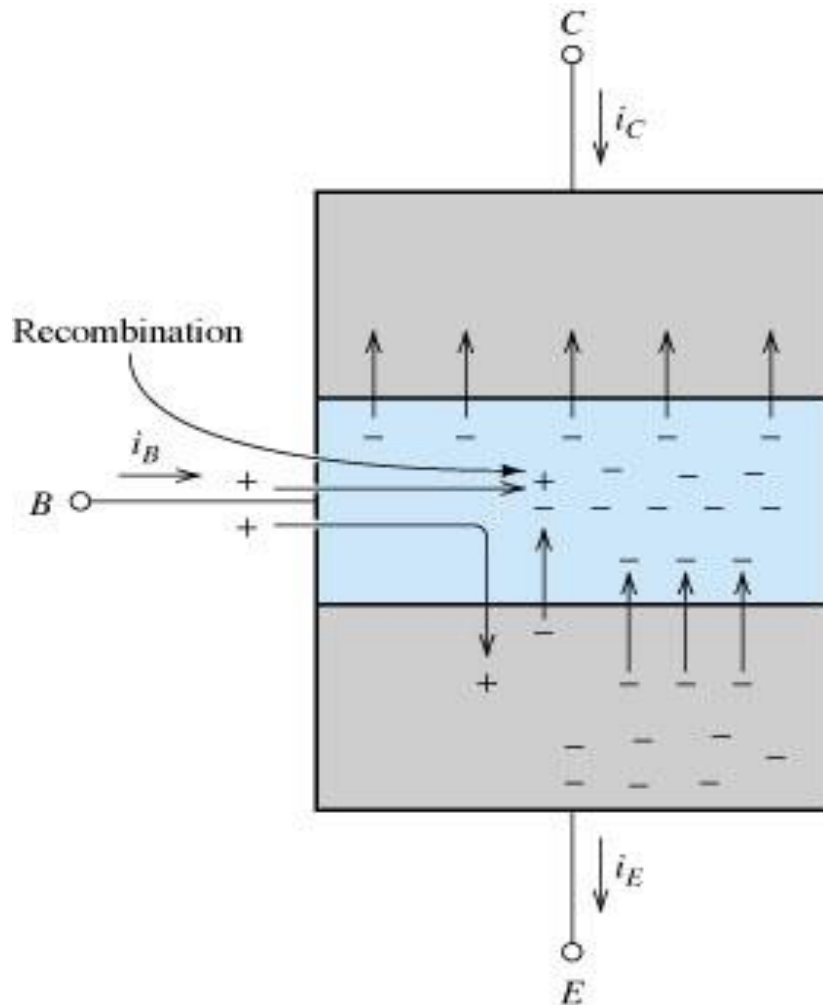
- Heavily doped emitter inject free electrons into the base
- Lightly doped base pass electrons on to the collector
- Collector collects or gathers electrons from the base



Biasing method – Emitter junction FB
Collector junction RB



Summary



- Forward biased emitter diode, forcing the free electrons in the emitter to enter the base
- Thin and lightly doped base diffuse electrons into collector
- Collector, through R_C and into the positive terminal of V_{CC}



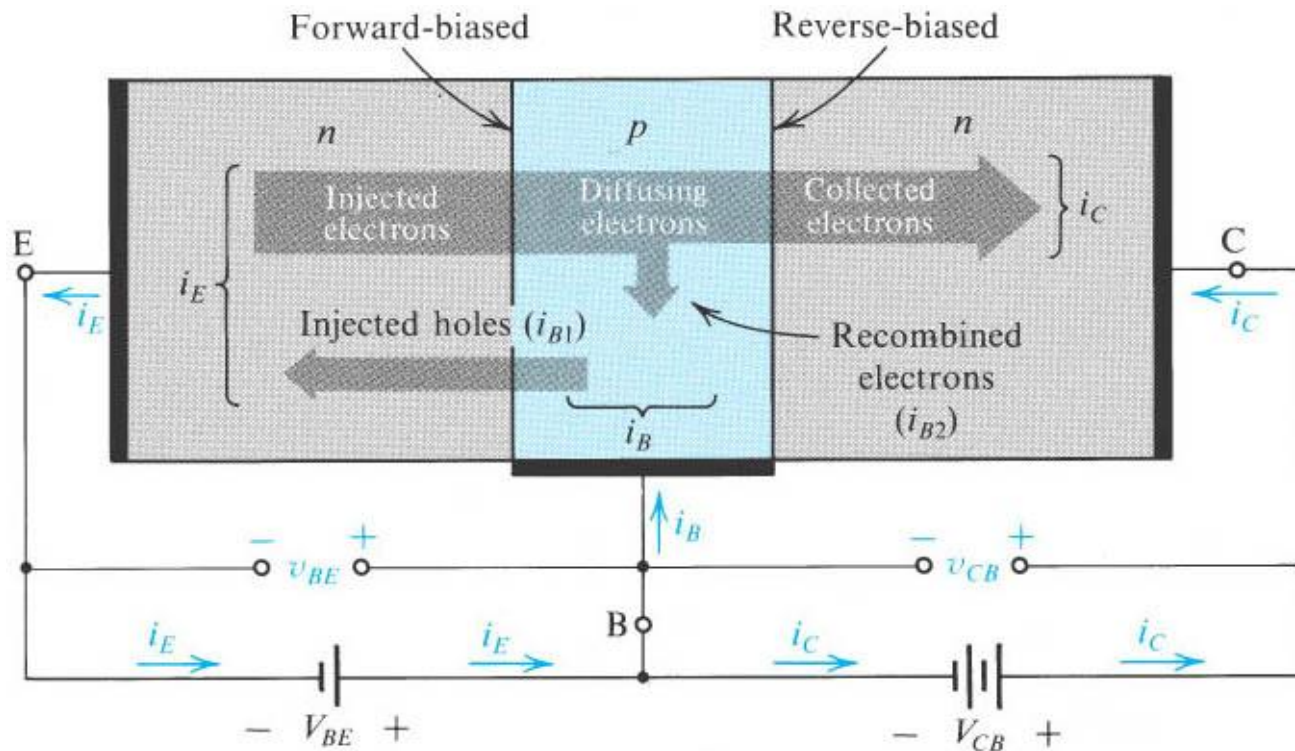
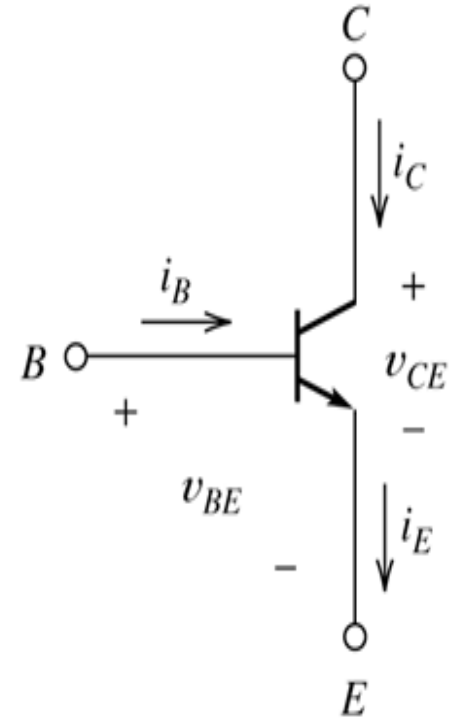


Figure 5.3 Current flow in an npn transistor biased to operate in the active mode. (Reverse current components due to drift of thermally generated minority carriers are not shown.)

Transistor Currents

- I_E – Largest emitter current
- Emitter electrons flow to the collector, $I_C \approx I_E$
- $I_B \leq 0.01 I_C$
- KCL, $I_E = I_C + I_B$



Circuit symbol



BJT α and β

- From the previous figure $I_E = I_B + I_C$
- Define $\alpha_{dc} = I_C / I_E$
- DC alpha is slightly less than 1
- Low power transistor $\alpha_{dc} > 0.99$ and High power transistor $\alpha_{dc} > 0.95$
- Define $\beta_{dc} = I_C / I_B$ - known as a current gain



BJT α and β

- Then $\beta_{dc} = I_C / (I_E - I_C) = \alpha_{dc} / (1 - \alpha_{dc})$
- **Assignment – Derive $\alpha_{dc} = \beta_{dc} / (1 + \beta_{dc})$**
- Then $I_C = \alpha_{dc} I_E$ & $I_B = (1 - \alpha_{dc}) I_E$

Solved Example 6.1, 6.2, 6.3

NPN BJT Current flow

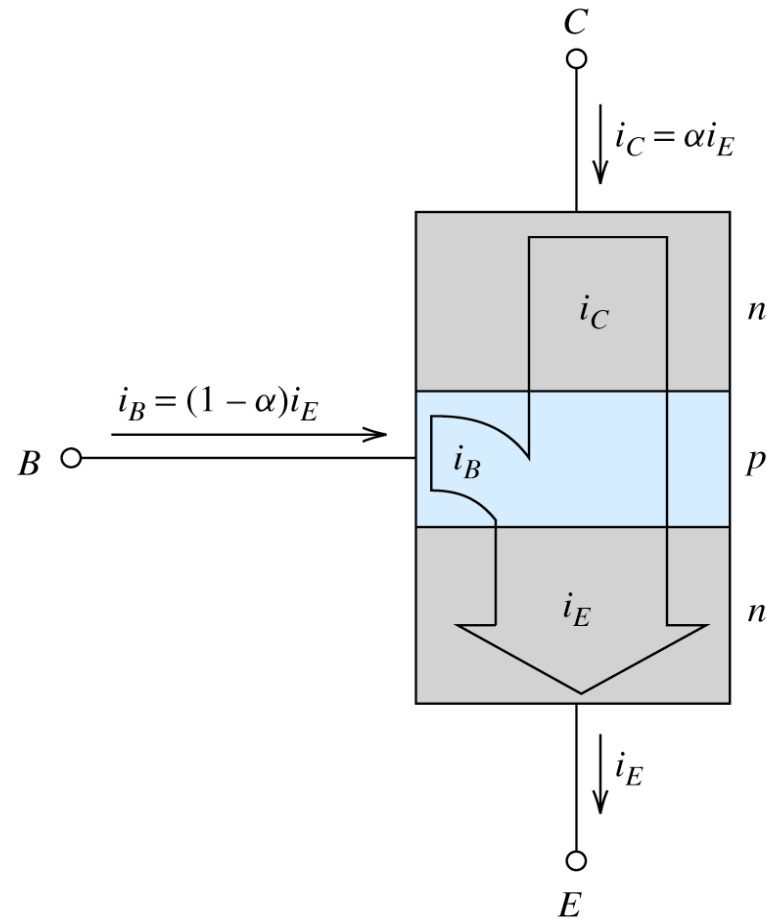
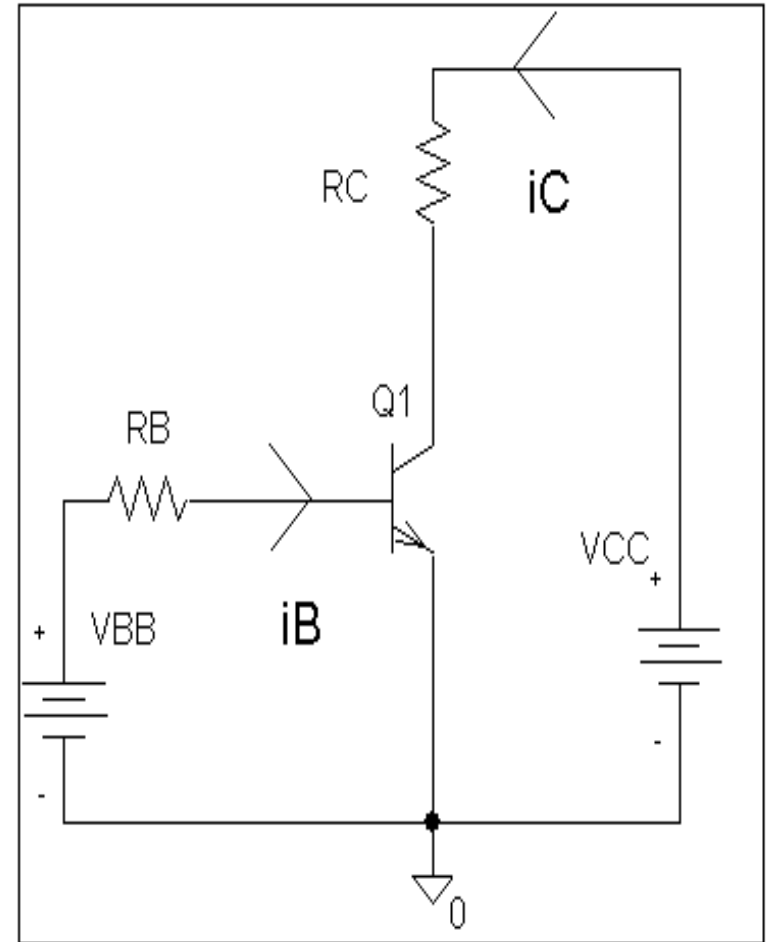


Figure 13.3 Only a small fraction of the emitter current flows into the base (provided that the collector–base junction is reverse biased and the base–emitter junction is forward biased).



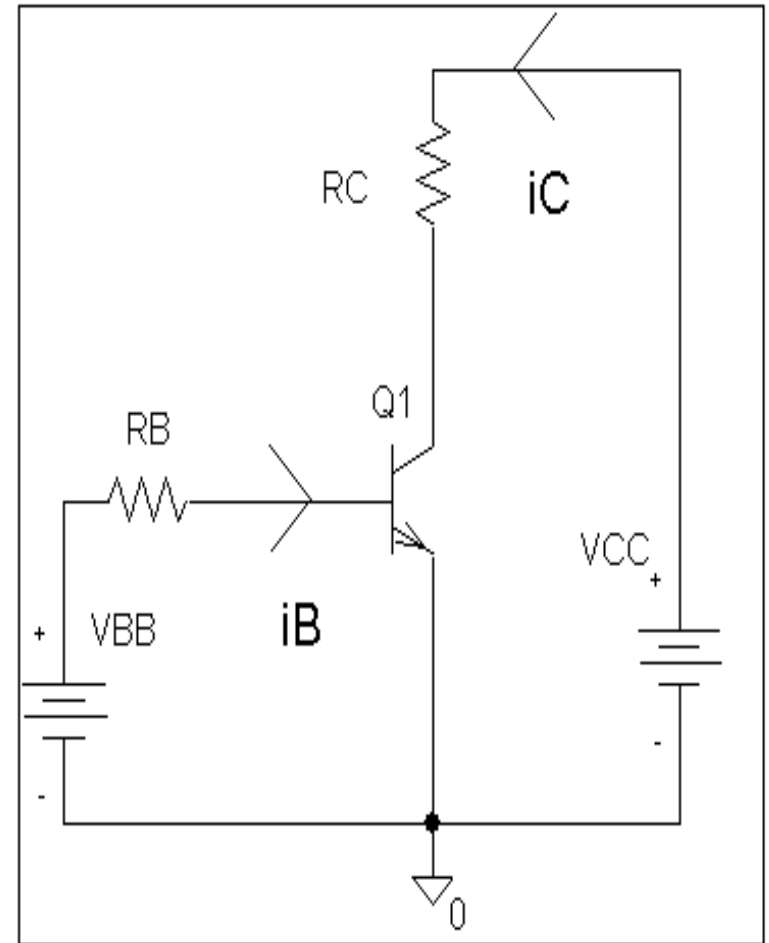
The CE connection

- CE, CC and CB
- CE because emitter is common to both V_{BB} and V_{CC}
- Left loop – Base loop
- Right loop – collector loop



The CE connection

- Base Loop, V_{BB} source and R_B – current limiting resistor
- Changing V_{BB} or R_B , change base current and I_B
Change than I_C change
- I_B controls I_C



Notation

Double Subscripts

- Voltage source – V_{BB} and V_{CC}
- V_{BE} – voltage between points B and E
- V_{CE} – voltage between points C and E

Single Subscripts

- Used for Node voltages
- V_B – voltage between base and ground
- V_C and V_E
- $V_{CE} = V_C - V_E$
- V_{CB} and V_{BE}

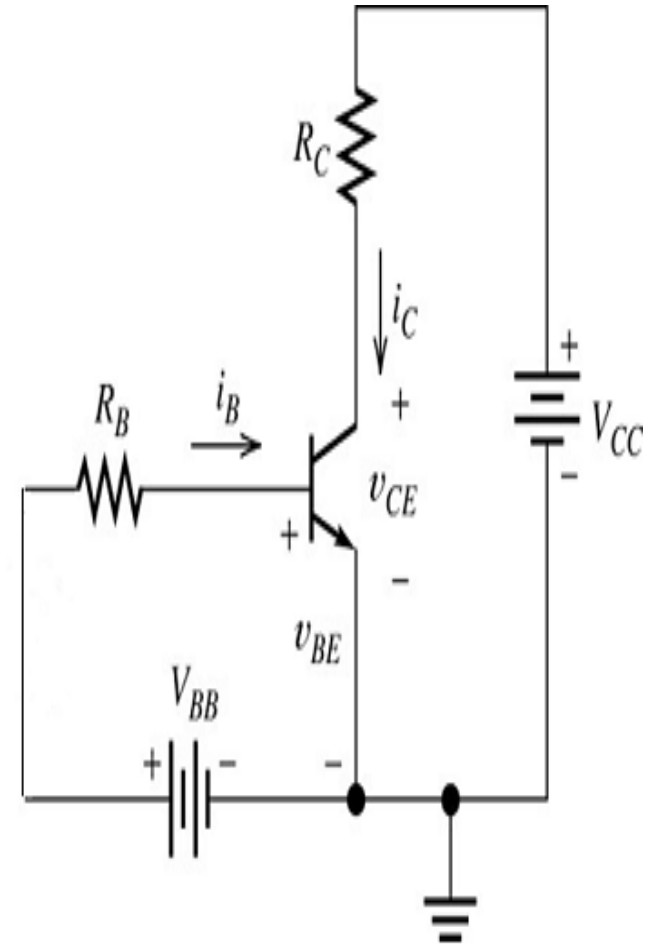
The Base Curve / Input Characteristics

- Graph I_B versus V_{BE}
- Like ordinary diode
- Ohm's law to Base loop

$$I_B = \frac{V_{BB} - V_{BE}}{R_B}$$

I

- Ideal diode $V_{BE} = 0$ and second app. $V_{BE} = 0.7 \text{ V}$

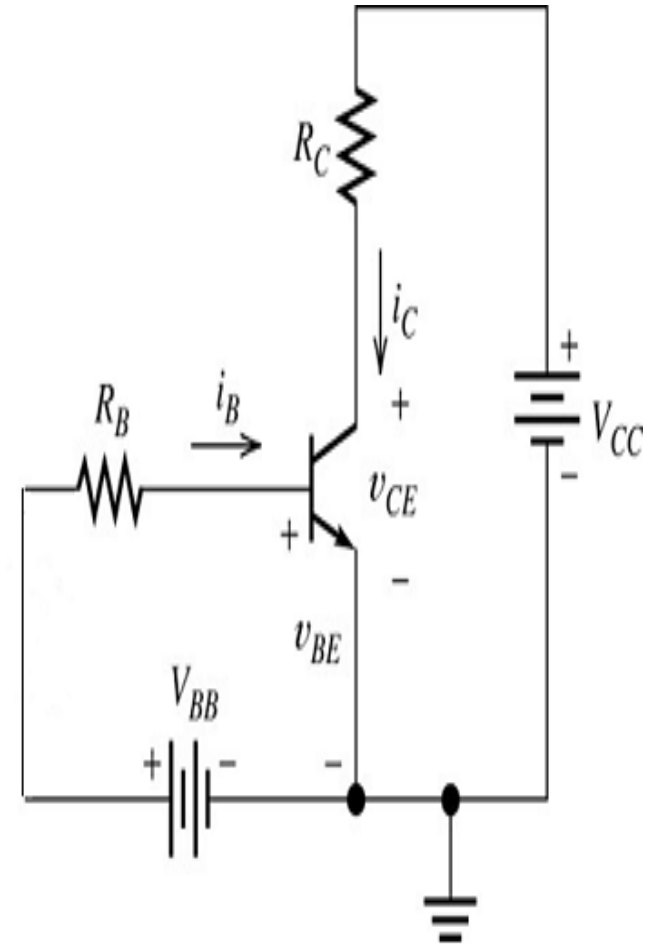


Collector Curve / output Characteristics

- Graph I_C versus V_{CE}
- Ohm's law to Collector loop

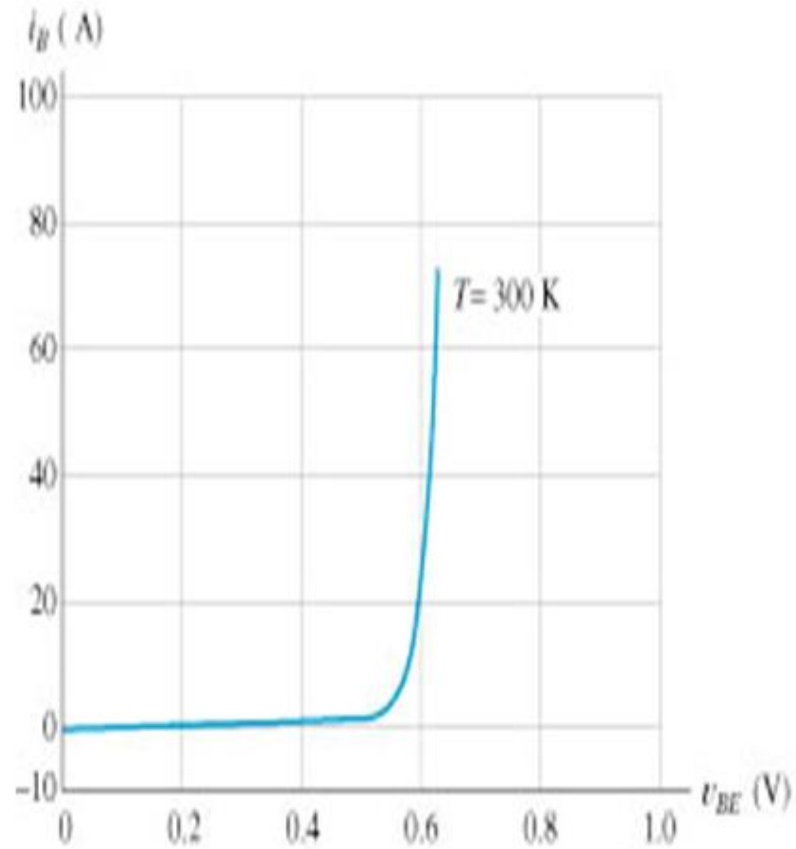
$$I_C = \frac{V_{CC} - V_{CE}}{R_C} \quad I$$

- Fixed value of based current, vary V_{CC} and measure I_C and V_{CE}

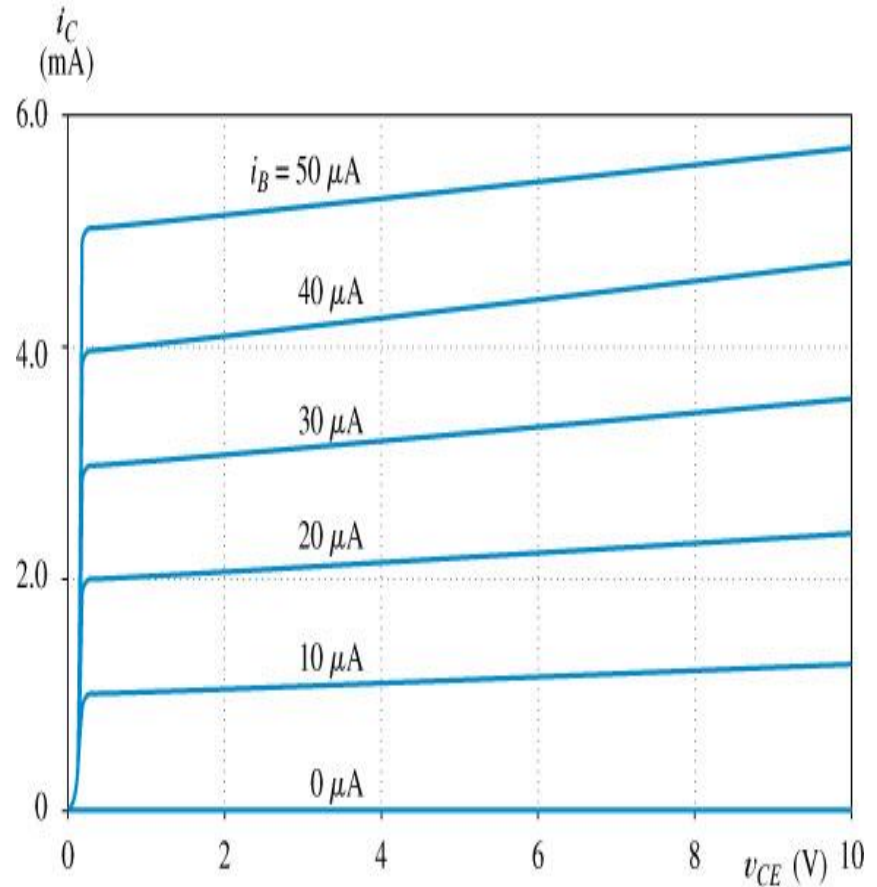


Transistor Characteristics

Input Characteristics



Output Characteristics



Active Region, Constant collector current

- After collector diode reverse biased, it collect all the electrons that reach its depletion layer
- Further increased V_{CE} cannot increased I_C
- Collector can collect only those free electrons that emitter injects
- $V_{CE} > V_{CE(max)}$, collector diode break down
- Power Dissipation $P_D = V_{CE} I_C$
- $P_D < P_{D(max)}$



Operating Region of Transistor

- **Active region**, middle region – normal operation of transistor

Emitter diode – FB and Collector diode – RB

- **Breakdown region** – transistor will be destroyed
- **Saturation region** – rising part of curve, V_{CE} between zero and few tenth of volt

Collector diode has insufficient positive voltage to collect all the free electrons injected into the base

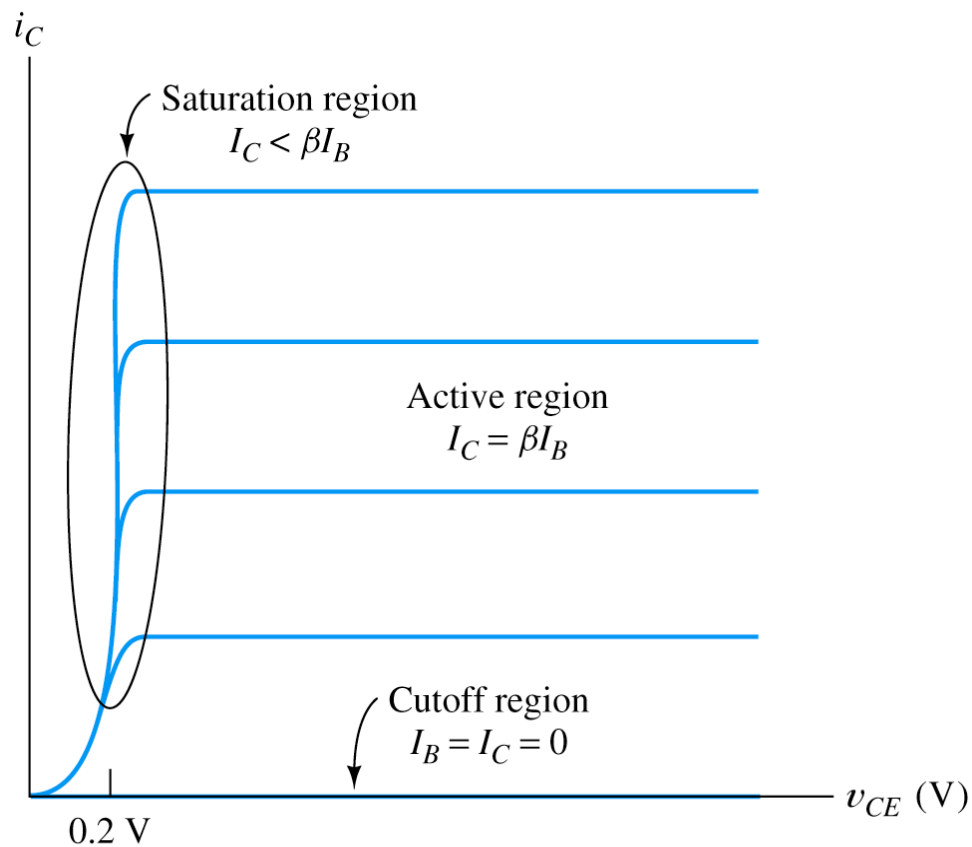
Operating Region of Transistor

- **Cut off region** – $I_B = 0$ but still small collector current

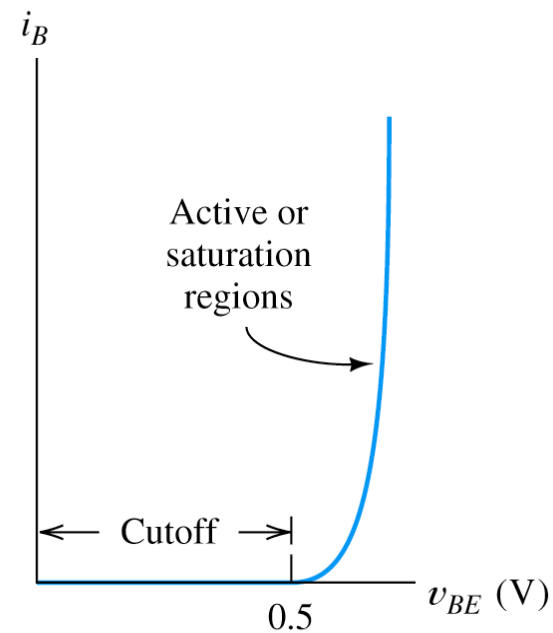
Because collector diode RB – Reverse minority carrier
+ Surface leakage current



BJT Operating Regions

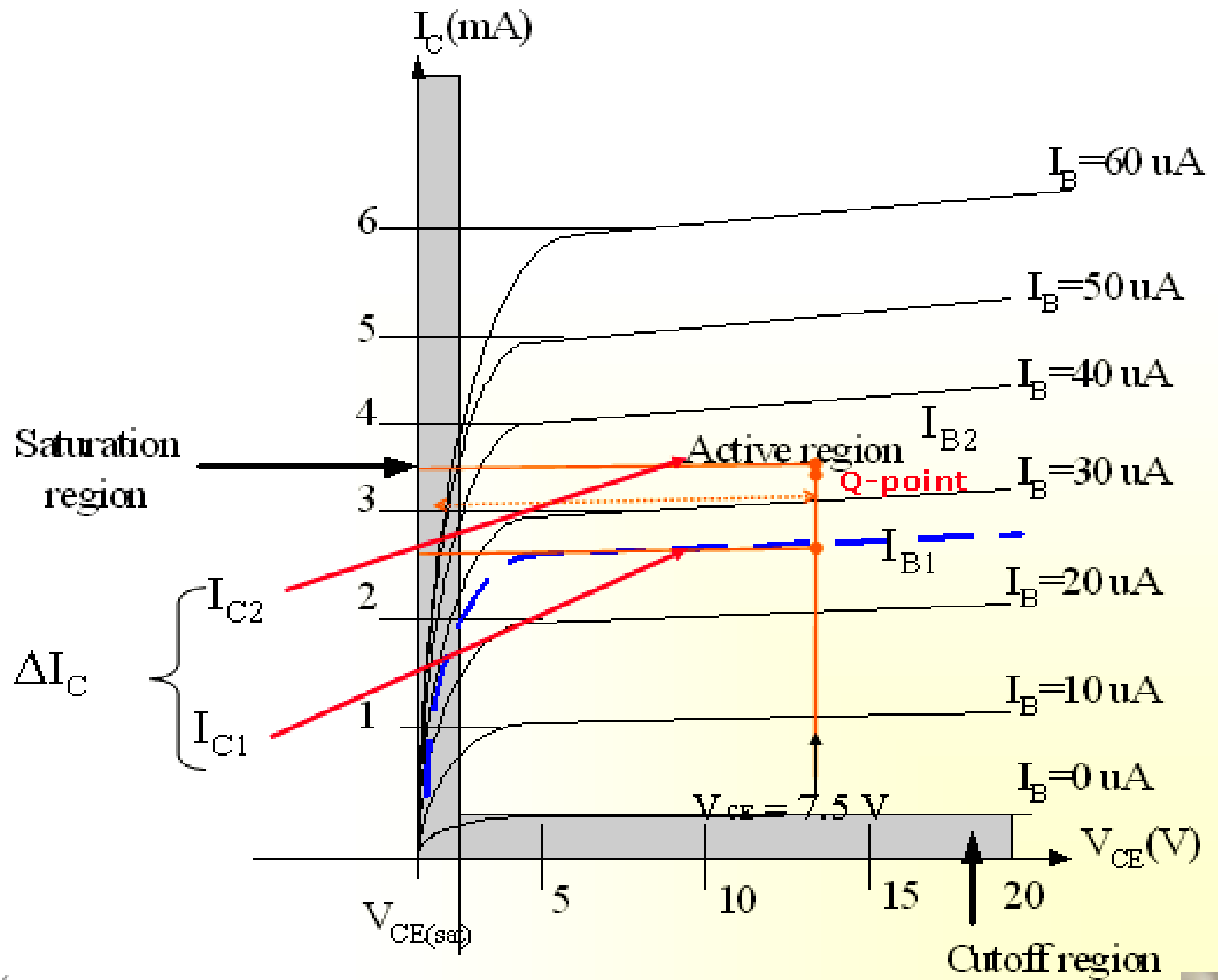


(a) Output characteristic

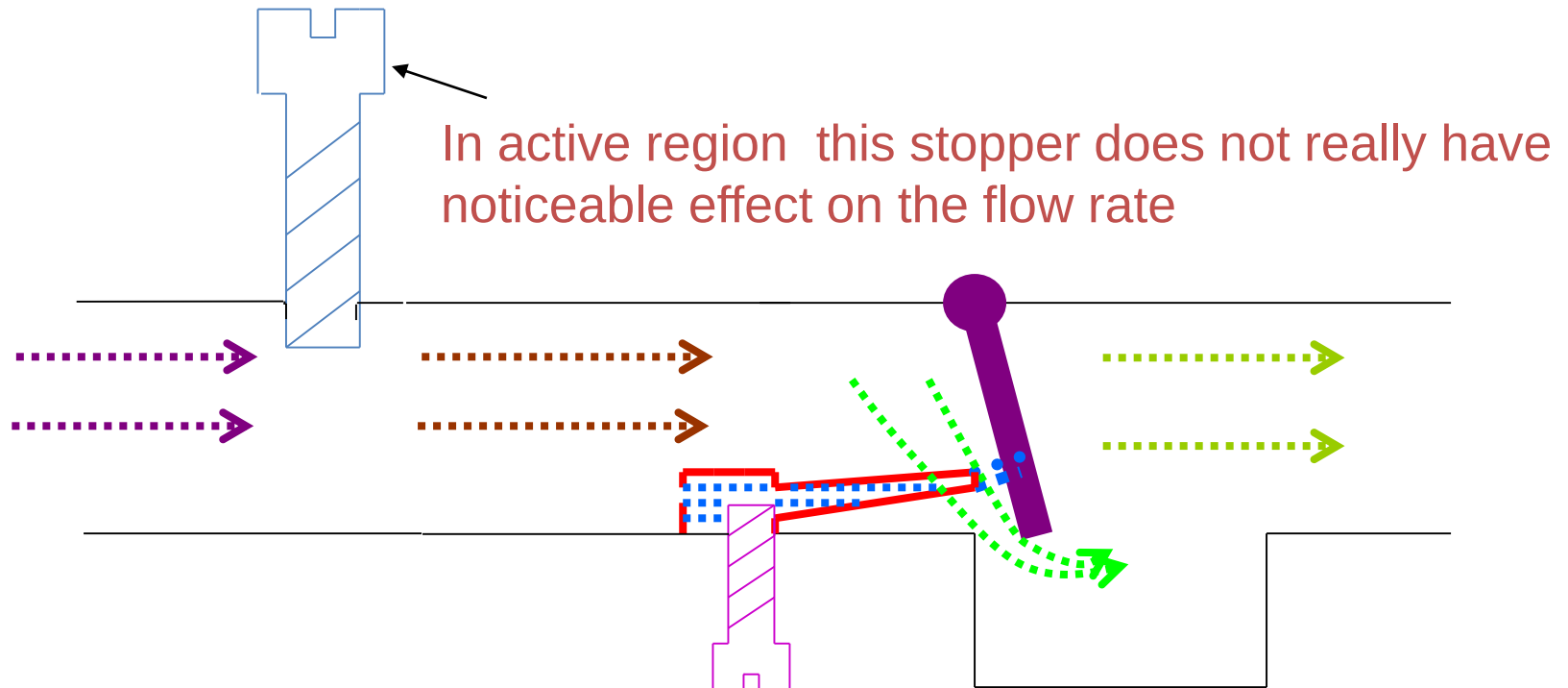


(b) Input characteristic

Figure 13.17 Regions of operation on the characteristics of an *npn* BJT.



Analogy with Transistor in Active Region: Fluid-jet operated Valve

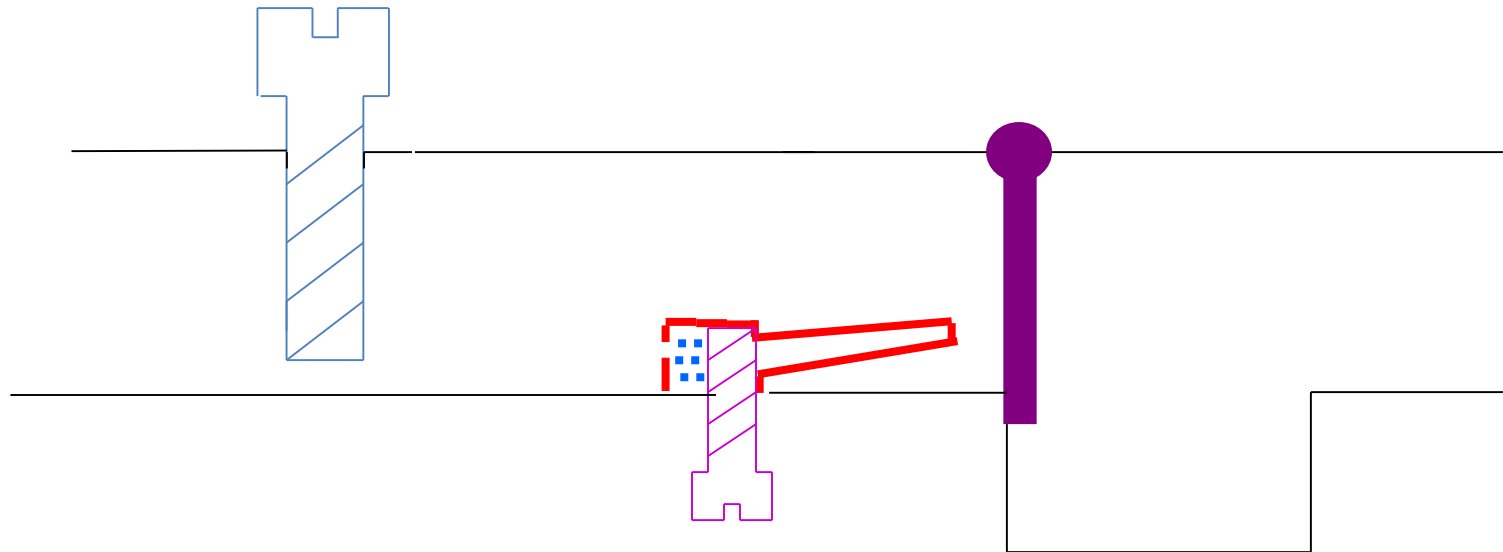


Emitter Base Junction – FB
Collector Base Junction – RB



Analogy with Transistor Cutoff

Fluid-jet operated Valve



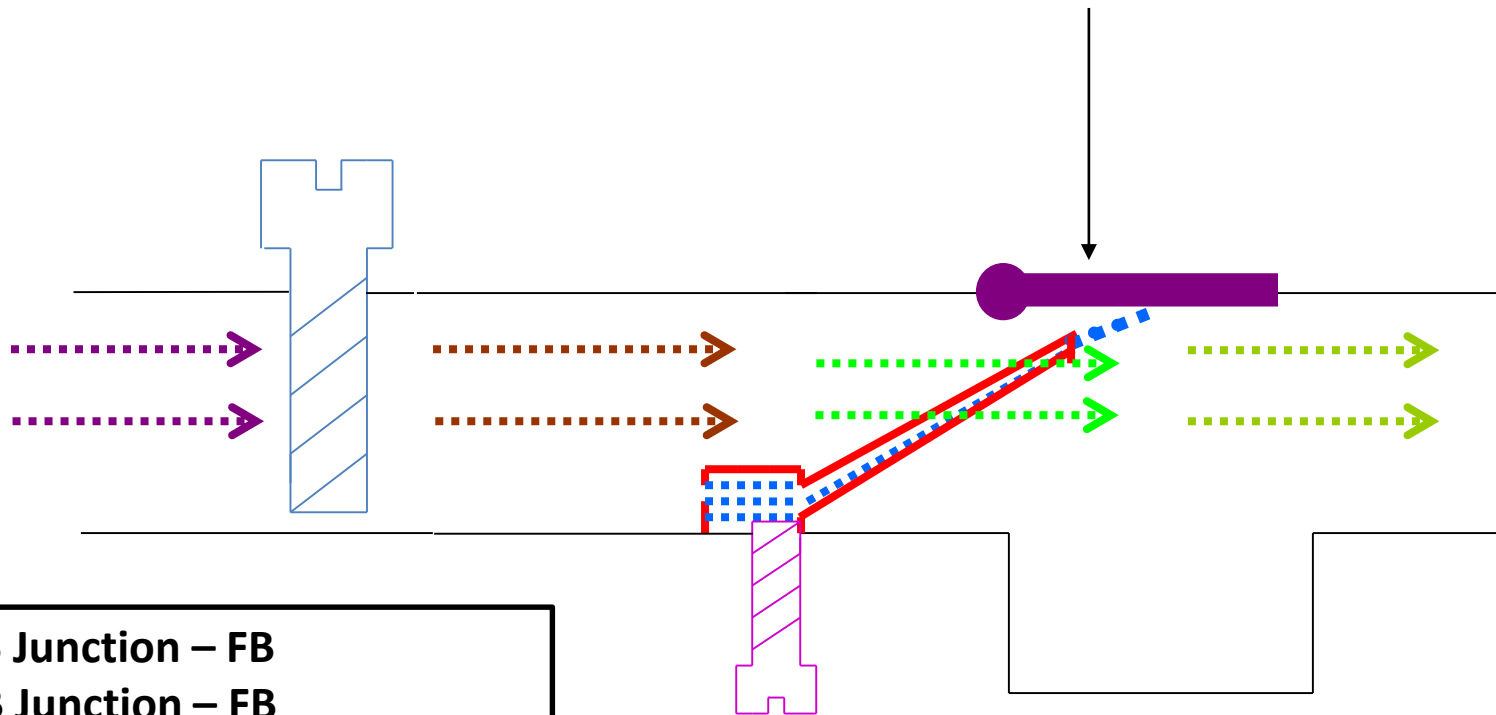
Emitter Base Junction – RB
Collector Base Junction – RB



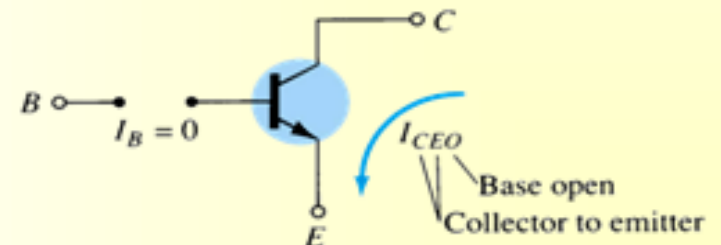
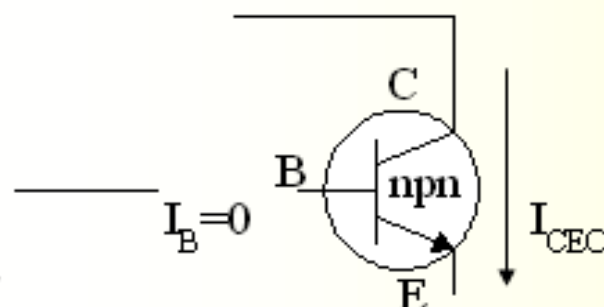
Analogy with Transistor Saturation

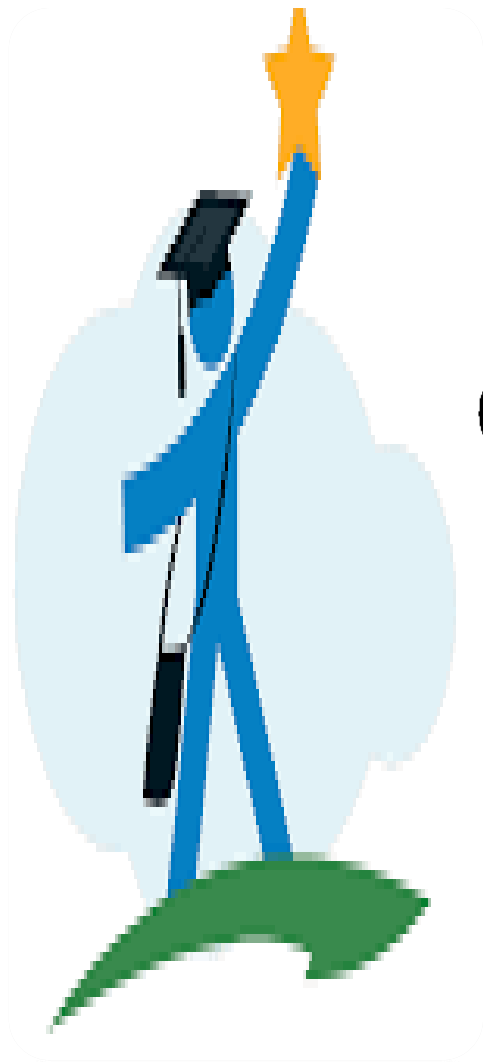
Fluid-jet operated Valve

The valve is wide open; changing valve position a little bit does not have much influence on the flow rate.



Active region	Saturation region	Cut-off region
• B-E junction is forward bias • C-B junction is reverse bias • can be employed for voltage, current and power amplification	• B-E and C-B junction is forward bias, thus the values of I_B and I_C is too big. • The value of V_{CE} is so small. • Suitable region when the transistor as a logic switch. • NOT and avoid this region when the transistor as an amplifier.	• region below $I_B=0\mu A$ is to be avoided if an undistorted o/p signal is required • B-E junction and C-B junction is reverse bias • $I_B=0$, I_C not zero, during this condition $I_C=I_{CEO}$ where is this current flow when B-E is reverse bias.





Thanks!

